Energy-Efficient Control of an Inverted Pendulum: PID vs. LQR vs. MPC vs. Reinforcement Learning

Amirreza Mohamadi and Reza Ghorbani

Technical report, team portfolio project, October 2026

Abstract

We compare four controllers for balancing an inverted pendulum on a cart with as little control energy as possible: a cascaded PID, a discrete LQR, a linear MPC solved with CasADi/qpOASES, and a Soft Actor-Critic (SAC) policy. All four run on the same validated nonlinear simulator with the same force limit, control rate and initial conditions, and all are tuned for the same cost $J = J_{\text{track}} + \rho E_u$, sweeping the energy weight ρ to trace energy–accuracy Pareto fronts. All controllers balanced the pendulum in every one of the 4×20 test runs. LQR forms the Pareto front in all four scenarios; the linear MPC reproduces LQR exactly whenever its constraints are inactive and only helps near the track limits; the grid-searched PID comes close to the front; the SAC policy needs 2–10$\times$ the tracking cost of LQR at equal energy in nominal regulation, but beats the LQR front under a push disturbance it was trained on.

1 Introduction

Every newton of actuator force drains the battery of a mobile robot, so a controller that reaches the same accuracy with less effort directly extends operating time. The inverted pendulum on a cart is the standard benchmark for comparing control approaches: it is unstable, nonlinear and underactuated, yet small enough to model by hand. We ask which of four controller families keeps the pendulum upright with the least control energy *at comparable accuracy*. Because a controller can always save energy by being less accurate, we report energy–accuracy Pareto fronts rather than a single ranking.

2 System Model

Equations of motion. With generalized coordinates $q = [x, \theta]^\top$ (cart position, pendulum angle, $\theta = 0$ upright, positive towards $+x$), a point-mass pendulum (mass m at distance l from the pivot) and no friction, the Lagrange method gives

$$\ddot{x} = \frac{u + ml\dot{\theta}^2 \sin\theta - mg\sin\theta\cos\theta}{M + m\sin^2\theta}, \tag{1}$$

$$\ddot{\theta} = \frac{(M+m)g\sin\theta - u\cos\theta - ml\dot{\theta}^2\sin\theta\cos\theta}{l\,(M + m\sin^2\theta)}. \tag{2}$$

The derivation was checked symbolically with SymPy. Parameters: $M = 1\,\text{kg}$, $m = 0.1\,\text{kg}$, $l = 0.5\,\text{m}$, $g = 9.81\,\text{m/s}^2$, $|u| \le 10\,\text{N}$, $|x| \le 2.4\,\text{m}$, $\Delta t = 0.02\,\text{s}$. Note that Gymnasium's `CartPole-v1` uses the same numbers but a uniform rod; our RL environment therefore wraps our own model.

Linearization. Around the upright equilibrium, with $s = [x, \dot{x}, \theta, \dot{\theta}]^\top$,

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+m)g}{Ml} & 0 \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{Ml} \end{bmatrix}, \tag{3}$$

i.e. $A_{24} = -0.981$, $A_{43} = 21.58$, $B = [0, 1, 0, -2]^\top$, with an unstable pole at $+4.65\,\text{rad/s}$ (the angle error doubles every $0.15\,\text{s}$). Zero-order-hold discretization at $50\,\text{Hz}$ gives (A_d, B_d).

Simulator and validation. The force is held constant over each $20\,\text{ms}$ period and the nonlinear model is integrated with 10 classical RK4 substeps. The simulator, not the controller, clips u to $\pm 10\,\text{N}$. Validation: (i) with $u = 0$ a pendulum tilted by 5° falls past 90° after $0.80\,\text{s}$; (ii) total energy drifts by only $1.4 \cdot 10^{-10}\,\text{J}$ over $10\,\text{s}$ and horizontal momentum stays zero; (iii) the RK4 error shrinks by more than $16\times$ per halving of the step; (iv) A, B equal the numerical Jacobian of (1)–(2); (v) linear and nonlinear responses differ by 0.02° at $\theta = 10^\circ$.

3 Controllers

All controllers share the interface $u = \text{controller}(s, t)$ and are tuned for the same continuous cost

$$J = \underbrace{\int_0^T (x - x_{\text{ref}})^2 + \theta^2 \, dt}_{J_{\text{track}}} + \rho \underbrace{\int_0^T u^2 \, dt}_{E_u}, \tag{4}$$

for $\rho \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$; $\rho = 0.01$ is the main setting.

PID (cascaded). An outer PD loop turns the cart error into a lean command, $\theta_{\text{ref}} = -(k_x e_x + k_v \dot{x})$, clipped to $\pm 10^\circ$ (to move right, the pendulum must first lean right). An inner PID on $e = \theta - \theta_{\text{ref}}$ gives $u = k_p e + I + k_d \dot{\theta}$ with back-calculation anti-windup ($\dot{I} = k_i e + (u_{\text{sat}} - u)/T_t$, $T_t = 0.1\,\text{s}$). After a manual start ($k_p = 30$, $k_d = 4$, $k_x = 0.1$, $k_v = 0.2$: oscillatory, $E_u = 106\,\text{N}^2\text{s}$ on the tuning set) the gains were chosen by an 800-point grid search on 10 tuning initial conditions. For $\rho = 0.01$ the grid search selects $k_p = 50$, $k_d = 11$, $k_x = 0.18$, $k_v = 0.15$, $k_i = 0$. With full state measurement, this cascade is a static state feedback $u = k_p\theta + k_d\dot{\theta} + k_p k_x e_x + k_p k_v \dot{x}$, and the selected gains are close to the LQR gains below. The integral term never improved the cost (no constant disturbance acts on this frictionless system).

LQR. Discrete LQR on (A_d, B_d) with $Q_d = Q\Delta t$, $Q = \text{diag}(1, 0, 1, 0)$, $R_d = \rho\Delta t$, the discrete equivalent of (4). For $\rho = 0.01$: $K = [-8.70, -8.20, -48.72, -10.60]$, $u = -K(s - s_{\text{ref}})$.

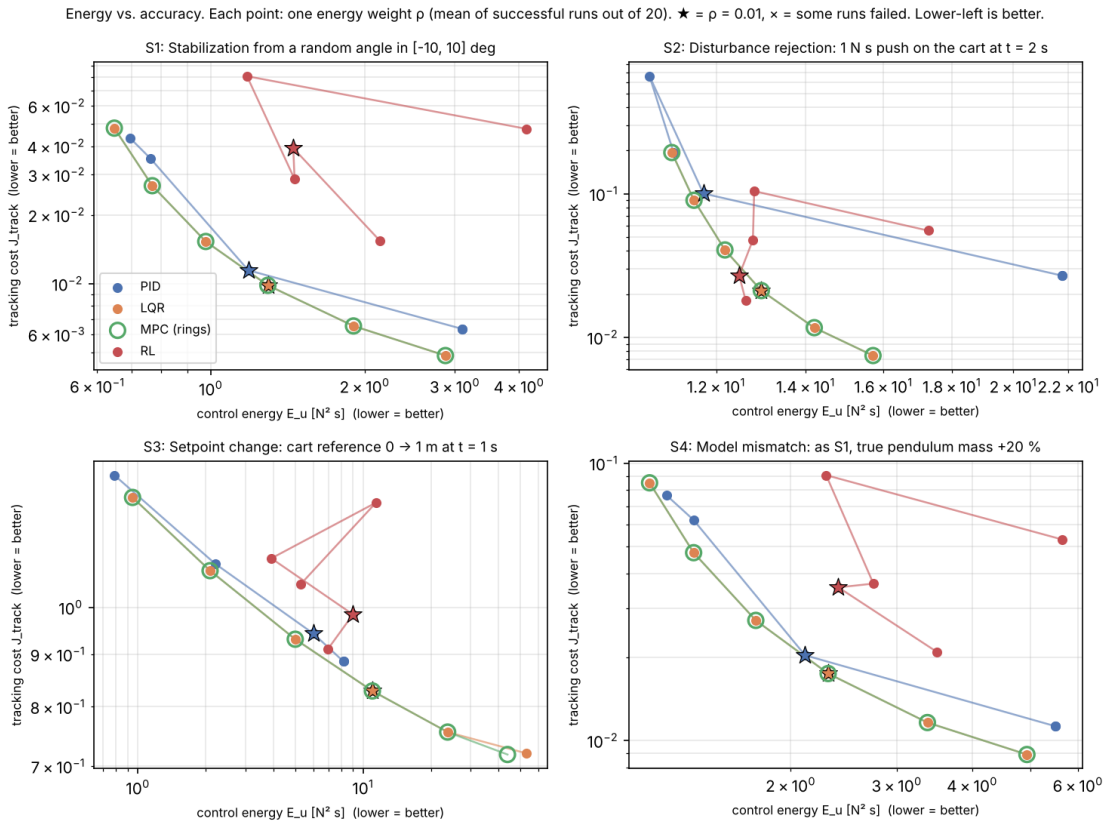


Figure 1: Energy–accuracy Pareto fronts. Each point is one energy weight ρ (mean of 20 runs); stars mark $\rho = 0.01$. MPC (rings) coincides with LQR. Lower-left is better.

MPC. Linear MPC with horizon N on the same model and cost, the Riccati matrix P of the LQR as terminal weight, $|u_k| \le 10\,\mathrm{N}$ and a soft track constraint $|x_k| \le 2.35\,\mathrm{m}$ (5 cm margin; slack penalized quadratically and linearly so the QP is always feasible). States are eliminated, leaving a dense QP with $N+1$ variables, solved by qpOASES through CasADi with warm starts. The energy term $\rho\,u^2$ is explicit in the cost. Only the current reference is used (no preview of future setpoints).

RL (SAC). A custom Gymnasium environment wraps our simulator; observation $[x - x_{\mathrm{ref}}, \dot{x}, \theta, \dot{\theta}]$, action $a \in [-1, 1]$, $u = 10a$. Reward (v2): $r = \max\!\big(0,\, 1 - \beta\,[(x - x_{\mathrm{ref}})^2 + \theta^2 + \rho u^2]\big)$ with $\beta = 5$, and -100 when the pendulum exceeds 30° or the cart leaves the track, so the energy penalty uses the same ρ as the other methods. Training randomizes the initial state ($|x_0 - x_{\mathrm{ref}}| \le 1\,\mathrm{m}$, $|\theta_0| \le 12°$, velocities ≤ 0.5) and, in half of the episodes, applies a random 0.5–1.2 N s push. SAC (Stable-Baselines3, 2×128 MLP, $\gamma = 0.995$, 150k steps, fixed seeds) was trained for $\rho \in \{0.001, 0.01, 0.03, 0.1, 0.3\}$, plus two extra seeds at $\rho = 0.01$. For each run, the best of the 50k, 100k and 150k-step checkpoints on the tuning set is used.

4 Experimental Setup

Four scenarios of 10 s each, 20 random initial conditions per scenario, drawn from a seed never used for tuning or training: **S1** start at $\theta_0 \sim \mathcal{U}(-10°, 10°)$, cart at rest; **S2** near-upright start ($|\theta_0| \le 2°$) and a 1 N s push (20 N for 50 ms) on the cart at $t = 2\,\mathrm{s}$; **S3** cart reference steps from 0 to 1 m at $t = 1\,\mathrm{s}$; **S4** as S1 but the true pendulum mass is 20 % higher than every controller assumes. A run fails if $|\theta| > 30°$ or $|x| > 2.4\,\mathrm{m}$. Metrics: $E_u = \int u^2 dt$, $E_{\mathrm{mech}} = \int |u\dot{x}| dt$, J_{track}, angle settling time t_s (into $\pm 1°$, measured from the push/step in S2/S3), maximum angle, cart overshoot (S3), success rate and controller computation time per step (Python, one CPU core). Means and standard deviations are over successful runs. Energy differences are tested against LQR with a paired Wilcoxon signed-rank test on the same initial conditions.

5 Results

Overall. All four controllers balanced the pendulum in all 4×20 evaluation runs at every energy weight (Table 1). Fig. 1 shows the energy–accuracy fronts; Table 2 compares each controller with LQR on the same initial conditions.

LQR and MPC define the front. In every scenario the LQR points form the lower-left boundary. MPC produces exactly the same trajectories as LQR. This is not a bug but a property of the design: with the Riccati matrix as terminal weight, an unconstrained linear MPC *is* the LQR. In the four scenarios the track is never approached ($|x| \le 1.03\,\mathrm{m}$) and the force limit is active only for a few samples during the push (LQR requests up to 11.9 N); there, the optimal constrained input equals the clipped LQR input. MPC costs about 0.3 ms per step against 0.003 ms for LQR, both far below the 20 ms period.

Table 1: Main results at $\rho = 0.01$: mean $\pm$ std over 20 initial conditions per scenario (all runs successful). t_s: settling time of the angle into $\pm 1^\circ$; t_c: controller computation time per step. Cart overshoot in S3: PID: 12.0 %, LQR: 2.7 %, MPC: 2.7 %, RL: 7.8 %.

Sc.	Ctrl.	E_u [N^{2}s]	E_{mech} [J]	J_{track}	t_s [s]	$\theta_{\max}$ [°]	Succ. [%]	t_c [ms]
S1	PID	1.18 ± 1.16	0.115 ± 0.113	0.011 ± 0.011	0.85 ± 0.72	4.0 ± 2.8	100	0.010
	LQR	1.29 ± 1.27	0.123 ± 0.121	0.010 ± 0.010	0.66 ± 0.51	4.0 ± 2.8	100	0.004
	MPC	1.29 ± 1.27	0.123 ± 0.121	0.010 ± 0.010	0.66 ± 0.51	4.0 ± 2.8	100	0.295
	RL	1.45 ± 1.45	0.108 ± 0.109	0.039 ± 0.040	1.84 ± 1.74	4.0 ± 2.8	100	0.137
S2	PID	11.74 ± 0.11	0.718 ± 0.012	0.101 ± 0.003	1.38 ± 0.01	6.6 ± 0.1	100	0.012
	LQR	12.97 ± 0.09	0.802 ± 0.009	0.021 ± 0.001	1.14 ± 0.01	6.7 ± 0.1	100	0.002
	MPC	12.97 ± 0.09	0.802 ± 0.009	0.021 ± 0.001	1.14 ± 0.01	6.7 ± 0.1	100	0.303
	RL	12.48 ± 0.16	0.678 ± 0.014	0.027 ± 0.002	2.22 ± 0.02	6.3 ± 0.2	100	0.171
S3	PID	6.01 ± 0.06	1.531 ± 0.018	0.943 ± 0.012	2.53 ± 0.01	8.9 ± 0.0	100	0.009
	LQR	11.00 ± 0.11	1.847 ± 0.006	0.829 ± 0.013	2.24 ± 0.00	12.0 ± 0.1	100	0.002
	MPC	11.00 ± 0.11	1.847 ± 0.006	0.829 ± 0.013	2.24 ± 0.00	12.0 ± 0.1	100	0.291
	RL	9.01 ± 0.79	1.513 ± 0.041	0.984 ± 0.030	3.21 ± 0.02	7.5 ± 0.1	100	0.119
S4	PID	2.11 ± 1.71	0.206 ± 0.167	0.020 ± 0.016	1.29 ± 0.89	5.5 ± 3.4	100	0.010
	LQR	2.31 ± 1.87	0.220 ± 0.179	0.017 ± 0.014	1.02 ± 0.70	5.5 ± 3.4	100	0.002
	MPC	2.31 ± 1.87	0.220 ± 0.179	0.017 ± 0.014	1.02 ± 0.70	5.5 ± 3.4	100	0.307
	RL	2.40 ± 1.98	0.192 ± 0.156	0.036 ± 0.032	1.69 ± 1.19	5.5 ± 3.4	100	0.108

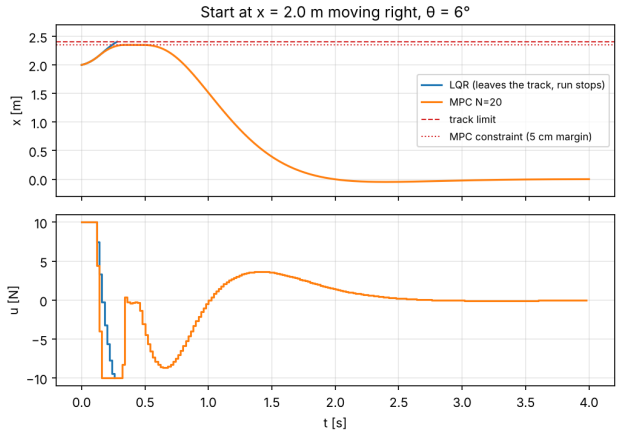


Figure 2: Start at $x = 2.0$ m moving right: LQR leaves the track, MPC brakes at its 2.35 m constraint and returns.

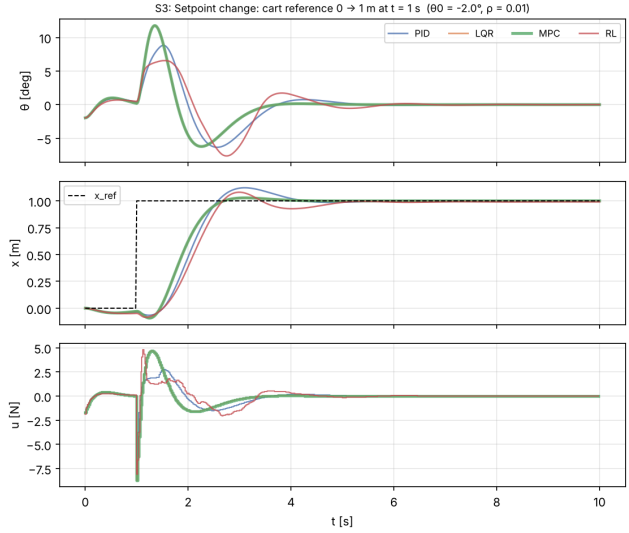


Figure 3: S3 (reference step at $t = 1$ s), $\rho = 0.01$. LQR lies under the MPC curve. PID and RL use less force but overshoot more.

PID. At $\rho = 0.01$ the tuned PID uses 8–46 % less energy than LQR (Table 2) but is less accurate in every scenario (tracking cost $\times 1.14$ to $\times 4.8$, overshoot 12 % vs. 2.7 % in S3, Fig. 3). At equal energy, its tracking cost is 1.01–1.42$\times$ that of the LQR front for $\rho = 0.01$, and up to 3.6$\times$ for extreme weights, where the coarse grid limits it. The saving is therefore bought with accuracy, not obtained for free.

RL. At $\rho = 0.01$ the SAC policy used 10 % *more* energy than LQR in S1 with a 3.9$\times$ higher tracking cost, and in S4 12 % more energy (not significant, $p = 0.37$) with 1.6$\times$ the tracking cost. At equal energy it needs 2.6–9.8$\times$ (S1) and 1.8–6.0$\times$ (S4) the tracking cost of the LQR front. S2 is the exception: here RL lies *beyond* the LQR front. The agent trained with $\rho = 0.001$ used 3 % less energy than LQR ($\rho = 0.01$) with 17 % lower tracking cost (paired Wilcoxon, $p < 0.001$). S2 is the one scenario whose disturbance type (a short push) was part of RL training, while LQR is optimal only for the undisturbed linear model. RL also delivered the least *mechanical* energy in all four scenarios (e.g. S2: 0.68 J vs. 0.80 J for LQR). The energy weight does not map monotonically to RL behavior: the $\rho = 0.3$ agent used more energy than the $\rho = 0.1$ agent, and its learning curve had not converged after 150k steps.

Robustness (S4). With a 20 % heavier pendulum, all controllers still succeeded in every run; the LQR tracking cost rose from 0.010 (S1) to 0.017 and energy from 1.29 to 2.31 N^{2}s, with the same ranking as in S1.

MPC horizon. Table 3 uses the tuning set. With the Riccati terminal weight, $N = 10$, 20 and 40 give identical closed loops; the solve time grows from 0.24 to 0.58 ms. Without a terminal weight, $N = 10$ (0.2 s) and $N = 20$ (0.4 s) are shorter than the dynamics need and fail in every run; $N = 40$ stabilizes but is 50 % less accurate.

Table 2: Paired comparison with LQR at $\rho = 0.01$: median ratio over the 20 initial conditions and Wilcoxon p-value for the energy difference.

Sc.	Ctrl.	E_u ratio	J_{track} ratio	p (Wilcoxon)
S1	PID	0.92	1.16	<0.001
S1	MPC	1.00	1.00	–
S1	RL	1.10	3.88	0.033
S2	PID	0.90	4.81	<0.001
S2	MPC	1.00	1.00	–
S2	RL	0.96	1.28	<0.001
S3	PID	0.54	1.14	<0.001
S3	MPC	1.00	1.00	–
S3	RL	0.84	1.18	<0.001
S4	PID	0.92	1.16	<0.001
S4	MPC	1.00	1.00	–
S4	RL	1.12	1.64	0.368

Table 3: MPC horizon study ($\rho = 0.01$, 10 tuning initial conditions).

Terminal cost	N	J_{track}	E_u	Fails	t_c [ms]
Riccati P	10	0.0827	2.74	0/10	0.244
Riccati P	20	0.0827	2.74	0/10	0.341
Riccati P	40	0.0827	2.74	0/10	0.581
none	10	–	–	10/10	0.284
none	20	–	–	10/10	0.384
none	40	0.1238	1.61	0/10	0.610
LQR (no MPC)	–	0.0827	2.74	0/10	0.002

When constraints matter. A supplementary test starts the cart at $x_0 \in \{1.4, \ldots, 2.0\}$ m moving towards the track end (36 starts, $|\theta_0| \leq 10°$). Success: LQR 14/36, MPC 19/36 (same for $N = 40$), PID 24/36, RL 30/36. MPC keeps the cart on the track where LQR leaves it (Fig. 2), but the linear prediction model limits how much it can gain. PID benefits from its 10° lean limit, and RL from having learned the track ends from the episode terminations.

RL reward design and variability. The first reward ($\beta = 1$, no clipping, $\gamma = 0.99$) produced a policy that balanced perfectly but stopped with the cart 19 cm from the reference: near the reference, the quadratic position penalty was too small to justify leaning the wrong way first, and a 2 s horizon was too short to see the benefit (S1 tracking cost 0.317). Scaling the cost ($\beta = 5$), clipping the reward at zero and $\gamma = 0.995$ reduced the offset to 1–3 cm. Checkpoints at 50k, 100k and 150k steps were compared on the tuning set; the final model was selected in 5 of 7 runs. Across three seeds at $\rho = 0.01$, the S1 tracking cost ranged from 0.019 to 0.039 and the S3 energy from 8.8 to 10.4 N^2s, a spread far larger than for the deterministic designs. Each agent took about 17 min of training on one CPU core.

6 Discussion

Which controller wins? For the question we posed, the most energy-efficient control at comparable accuracy, *LQR* wins: it sits on the Pareto front in every scenario, is the simplest to design and the cheapest to run. This is expected, since our shared cost is quadratic and the pendulum stays in a nearly linear range ($|\theta| \leq 12°$), so LQR is the optimal solution of exactly this problem for the undisturbed model. MPC matches LQR and adds value only where constraints bind, as near the track end. The tuned PID is a good baseline whose structure is equivalent to state feedback, but its energy savings come with lower accuracy. RL needs no model at run time and handled pushes and track ends best, but needed a reward redesign (Fig. 4), shows large seed-to-seed variation and does not reach the front in nominal regulation.

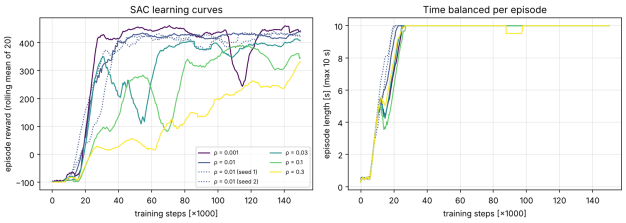


Figure 4: SAC learning curves (reward v2). All agents balance for the full 10 s after about 25k steps; larger energy weights learn more slowly.

Limitations. Point-mass pendulum, no friction, no sensor noise or delay, full state measurement; MPC is linear (nonlinear MPC was not implemented); RL used a single training budget of 150k steps and was the only method trained with pushes, which favors it in S2; computation times are Python wall times on one core; the fall threshold of 30° is our choice.

7 Conclusion

On a validated nonlinear cart-pole simulator with a common cost and identical test conditions, LQR provides the best energy–accuracy trade-off; linear MPC with an LQR terminal weight reproduces it and is only worth its 100× higher computation when constraints are active; a grid-searched PID comes close; and SAC is competitive under disturbances it was trained on but otherwise needs 2–10× the tracking error for the same energy. Natural next steps are nonlinear MPC, sensor noise and delay, and deployment on embedded hardware.

Reproducibility. All results follow from the repository with fixed seeds (evaluation seed 2026, tuning seed 7, RL seeds 0–2); see `README.md` for the commands. AI coding tools were used for help with parts of the code.

References

[1] A. G. Barto, R. S. Sutton, C. W. Anderson, "Neuronlike adaptive elements that can solve difficult learning control problems," *IEEE Trans. SMC*, 1983.

[2] K. J. Åström, R. M. Murray, *Feedback Systems*, 2nd ed., Princeton Univ. Press, 2021.

[3] J. B. Rawlings, D. Q. Mayne, M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., Nob Hill, 2017.

[4] J. A. E. Andersson et al., "CasADi: a software framework for nonlinear optimization and optimal control," *Math. Prog. Comp.*, 2019.

[5] H. J. Ferreau et al., “qpOASES: a parametric active-set algorithm for quadratic programming,” *Math. Prog. Comp.*, 2014.

[6] R. S. Sutton, A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

[7] T. Haarnoja et al., “Soft actor-critic: off-policy maximum entropy deep RL with a stochastic actor,” ICML, 2018.

[8] A. Raffin et al., “Stable-Baselines3: reliable reinforcement learning implementations,” *JMLR*, 2021.